SURGE × OPENCLAW HACKATHON 2026

SuperPage

The x402 Payment Skill for the Agent Internet

An OpenClaw skill that lets autonomous AI agents discover, preview, confirm, and pay for digital resources and physical products using on-chain USDC micro-payments on Base Sepolia.

x402 Protocol · OpenClaw · Base Sepolia · MCP · A2A / AP2 / ERC-8004

The Problem

AI Agents Are Locked Out of Commerce

> *Agents can research, write, and plan — but they can't pay for anything.*

Premium APIs, digital content, and e-commerce are all trapped behind human checkout flows. There's no standard way for an agent to discover a resource, see its price, confirm with the user, and pay.

WHAT AGENTS CAN DO	WHAT THEY CAN'T
Write code	Pay for cloud compute
Research products	Purchase products
Analyze data	Subscribe to data APIs
Plan trips	Book & pay for hotels
Compare prices	Complete checkout

The Solution

A Complete Agent Payment Workflow

SuperPage gives any AI agent the ability to discover, evaluate, pay for, and access resources — all autonomously with user confirmation.

1. Discover
list-resources / search

→

2. Preview
price check (free)

→

3. Confirm
"Buy for $0.50?"

→

4. Pay
USDC on Base

→

5. Deliver
content + receipt

Works across **three surfaces**: Telegram (OpenClaw), Claude Desktop (MCP), and CLI — all powered by the same `superpage-x402.js` engine.

Architecture

Full-Stack AI-Native Commerce

Agent Surfaces

One Skill, Three Surfaces

Telegram

OpenClaw skill powering **@HeySuperioBot**. Agent previews price, asks user to confirm, pays on-chain, delivers content.

"Buy the Weather API"
→ "It costs $0.50 USDC. Proceed?"
→ "Yes"
→ Paid! Here's the data...

Claude Desktop (MCP)

12 autonomous tools via Model Context Protocol. Claude can browse, pay, and access resources natively.

- `x402_list_resources`
- `x402_preview / x402_request`
- `x402_buy (Shopify)`
- `x402_wallet / x402_send`

CLI Agent

Multi-LLM standalone agent via Vercel AI SDK. Supports Anthropic, OpenAI, and Google models.

```
$ pnpm agent "buy weather API"

Found Weather API for $0.50 USDC.
Confirm? yes
Paid 0.50 USDC — tx: 0x01e5...
Here's the data: { temp: 72 }
```

x402 Protocol

HTTP 402 Payment Flow

1. **Agent Requests Resource**
   `GET /x402/resource/weather-api`

2. **Server Returns 402**
   Payment requirements: amount, currency, recipient, chainId

3. **Preview Mode (Stops Here)**
   Returns price to user — "Buy for $0.50 USDC?"

4. **User Confirms → Agent Pays**
   USDC transfer on Base Sepolia, ~2s confirmation

5. **Retry with Proof → Content Delivered**
   X-PAYMENT header with tx hash, on-chain verification, resource served

```
// 1. Request resource
GET /x402/resource/weather-api

// 2. Server responds: 402
{
  "amount": "0.50",
  "currency": "USDC",
  "recipient": "0x...",
  "chainId": 84532
}

// 3. Agent pays USDC on Base
await sendUSDC(recipient, "0.50")
// tx: 0x9eab1d...

// 4. Retry with payment proof
GET /x402/resource/weather-api
Headers: {
  "X-PAYMENT": "0x9eab1d..."
}

// 5. 200 OK — content delivered
{ "temperature": 72, ... }
```

Protocol Stack

Five Protocols, One Platform

SuperPage is the first platform to combine all five agent protocols into a single commerce infrastructure.

x402
HTTP 402 payment-gated resources. The payment standard for the agent internet.

A2A
Agent-to-agent discovery via /.well-known/agent.json. JSON-RPC 2.0.

AP2
Google's Agent Payments. Intent → Cart → Payment → Receipt mandate flow.

MCP
Model Context Protocol. 12 tools for Claude Desktop native integration.

ERC-8004
On-chain agent identity, reputation scoring, and trust validation.

AP2 Shopping Flow (Google Agent Payments Protocol)

IntentMandate — What to buy → **CartMandate** — Itemized cart → **PaymentMandate** — On-chain tx → **PaymentReceipt** — Verified settlement

Platform Features

Two-Sided AI-Native Marketplace

For Creators & Merchants

- → List digital resources: APIs, datasets, guides, articles
- → Connect Shopify store in 3 clicks — products auto-sync
- → Set prices in USDC — receive payments directly on-chain
- → Creator dashboard with analytics and order management
- → Public profiles with tipping support
- → Earn from both humans and AI agents

26+ Digital Resources

12 MCP Tools

5 Protocols

Shopify Integration

```
// Agent browses store products
browse-products {
  "storeId": "shopify/x402-store"
}

// Returns products with variants
[{
  "title": "Example T-Shirt",
  "price": "0.01 USDC",
  "variantId": "gid://shopify/..."
}]

// Agent completes full checkout
buy {
  "storeId": "shopify/x402-store",
  "items": [{
    "productId": "gid://...",
    "quantity": 1
  }],
  "shippingAddress": { ... }
}
// → Order created + USDC paid
```

Resource Types

API — $0.01-$1 | File — $0.50-$50 | Article — $0.10-$10 | Shopify — varies

Safety & Security

Trust Built Into Every Layer

🔒

Spending Caps

MAX_AUTO_PAYMENT limits per-transaction spend. Default $10 USDC max.

👁️

Preview Before Pay

Agent always shows price and asks for user confirmation before any payment.

🔑

Non-Custodial

Agent controls its own private key. No third party holds funds.

⛓️

On-Chain Verification

Every payment cryptographically verified against Base Sepolia before content is served.

⏱️

Time-Bounded

15-minute payment windows prevent stale or replayed transactions.

📋

Full Audit Trail

Transaction hash, amount, timestamp, recipient — everything logged and verifiable.

```
// Every transaction receipt includes:
{
  "txHash":    "0x9eab1d...",
  "amount":    "0.50 USDC",
  "to":        "0x5163...",
  "chainId":   84532,
  "verified":  true,
  "timestamp": 1709312400
}
```

Technology

Monorepo Architecture

LAYER	STACK
Frontend	Next.js 16, React 19, Tailwind 4
Backend	Express, TypeScript, MongoDB
MCP Client	Node.js, viem, JSON-RPC
AI Agent	Vercel AI SDK (multi-LLM)
SDK	@super-x402/sdk (payment verification)
Blockchain	Base Sepolia, viem, USDC
E-Commerce	Shopify Admin API
Identity	ERC-8004
Wallet	RainbowKit, wagmi
Agent Runtime	OpenClaw (Telegram gateway)

```
superpage/
├── packages/
│   ├── frontend/
│   │   Next.js 16 + React 19 + Tailwind 4
│   │
│   ├── backend/
│   │   Express + MongoDB + A2A + AP2 + x402
│   │
│   ├── mcp-client/
│   │   MCP server + CLI (superpage-x402.js)
│   │
│   ├── ai-agent/
│   │   Standalone AI agent (multi-LLM)
│   │
│   ├── x402-sdk-eth/
│   │   Payment verification SDK
│   │
│   └── contracts/
│       Smart contract ABIs
│
├── dev.sh
└── package.json  pnpm workspace
```

Hackathon Tracks

Track Alignment

Track 5: Autonomous Payments & Monetized Skills

x402-integrated OpenClaw skill charging USDC per resource. Preview + confirm flow, micro-payments ($0.01–$5), 12 MCP tools, direct creator revenue, reusable skill any OpenClaw agent can install.

Track 1: Agent Execution

Autonomous Shopify shopping, wallet management, full checkout pipeline with shipping.

Track 3: Developer Infrastructure

x402 SDK, MCP server, A2A protocol, AP2 implementation — reusable building blocks.

Track 4: Agent-to-Agent

A2A discovery, AP2 mandate-based shopping, ERC-8004 identity and reputation.

What Makes Us Different

Full Platform

Not a demo — real marketplace with 26+ resources and Shopify integration

Multi-Surface

Telegram + Claude Desktop + CLI — same engine, three interfaces

5 Protocols

x402, A2A, AP2, MCP, ERC-8004 — most comprehensive protocol stack

Live Demo

See It In Action

Telegram (@HeySuperioBot)

1. **"What resources are available?"**
   Lists all 26+ resources with names, descriptions, prices

2. **"Buy the Weather API"**
   Previews price → asks "Buy for $0.50?" → waits for yes

3. **"Yes"**
   Pays USDC on-chain → delivers content → shows tx receipt

4. **"Buy a t-shirt from the Shopify store"**
   Browses products → selects variant → full checkout with shipping

Try It Now

Live Demo
superpage.fun
`Browse marketplace, connect wallet, mint test USDC`

Telegram Bot
@HeySuperioBot
`Message the bot → ask to buy a resource`

Agent Discovery
/.well-known/agent.json
`A2A agent card with skills & extensions`

GitHub
AIProjects402/superpage
`Full source code — MIT license`

Network Details

Network	Base Sepolia
Chain ID	84532
Currency	USDC (6 decimals)
Block Time	~2 seconds

SuperPage

The x402 Payment Skill for the Agent Internet

Where AI agents and humans coexist as buyers and sellers.

superpage.fun | @HeySuperioBot | GitHub

Built for **SURGE × OpenClaw Hackathon** — February 2026

x402 | A2A | AP2 | MCP | ERC-8004